

Power and Cooling -- Guided Notes

Unit 2.4 | CompTIA Network+ N10-009 Obj. Core 1 3.5 | N10-008

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

The ATX PSU provides three main DC voltage rails. The _____ V rail powers the CPU and GPU and is the most heavily loaded rail. The _____ V rail powers older drives and some motherboard logic. The _____ V rail powers RAM and low-voltage logic. The wattage rating of the PSU represents its maximum _____ output.

Question 2

The 80 PLUS certification rates PSU _____. An 80 PLUS Gold PSU achieves approximately _____ % efficiency at 50% load, meaning _____ % of the power drawn from the wall is wasted as heat. A more efficient PSU produces less _____ and reduces electricity costs.

Question 3

When selecting a PSU wattage, technicians add up the TDP of all components and add _____ to _____ % as a headroom buffer. This prevents the PSU from running near _____ % capacity, which reduces efficiency and component lifespan. A system with a 300W total TDP should use a PSU rated at least _____ W.

Question 4

The EPS _____ -pin connector (sometimes called the CPU power connector) delivers +12V power directly to the _____ voltage regulator module on the motherboard. Without this connector plugged in, high-end CPUs may _____ POST or operate at reduced power.

Question 5

Thermal paste (also called thermal compound) is applied between the CPU and heatsink to fill _____ air gaps in the metal surfaces. It improves _____ transfer from the CPU to the heatsink. The recommended application amount is approximately _____ -sized -- too much can _____ onto the motherboard and cause shorts.

Question 6

An AIO (All-In-One) liquid cooler is a _____ cooling system. It consists of a _____ block that mounts on the CPU, connected by tubes to a _____ where heat is dissipated. AIO coolers generally provide _____ thermal performance than air coolers for high-TDP processors, but require more _____ mounting space.

Question 7

Case fans are labeled with a direction indicator. The _____ side of the fan is the exhaust side. For optimal airflow, intake fans should be mounted at the _____ and front, while exhaust fans go at the _____ and top. A system with more _____ fans than exhaust fans creates positive pressure, which reduces dust buildup.

Question 8 -- Real World Application

REAL WORLD: A student builds a PC and notices that after 10 minutes of gaming, the system stutters, frame rates drop sharply, and then performance returns to normal after a minute. No BSOD occurs.

Explain the most likely cause, identify the technical term for this behavior, and describe two corrective actions the technician should take.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/2-4-power-and-cooling/2-4-power-and-cooling-answer-key.pdf
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